Foundations of Project Management for Innovative Outcomes

Learning Outcomes:

LO1. Understanding the project performance dimensions

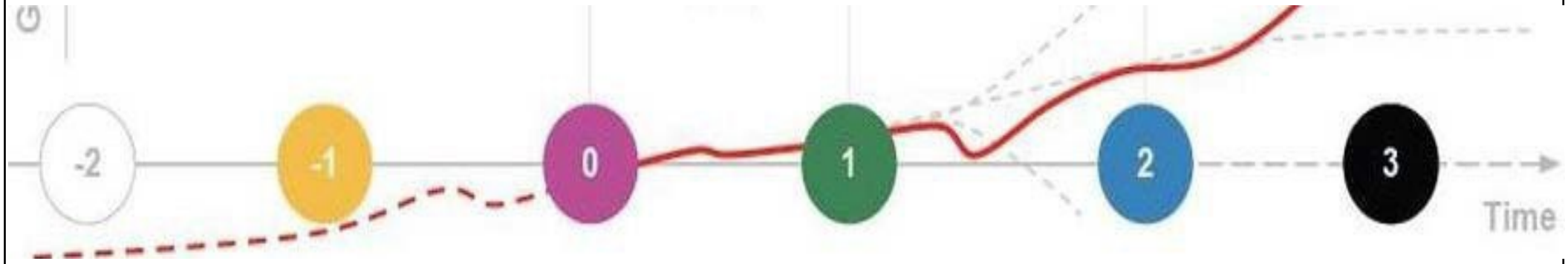
LO2. Familiarization with the Project Governance – Life Cycle Framework

LO3: An Introduction to Project Typology based on the innovation landscape

LO4: Appreciating the difference between predictive and adaptive project management and the decision frameworks to select the right project management approach.

CRICOS 00111D
TOID 3059

PRM60002 Framework
Project Management for Innovative Outcomes

Entrepreneurship	Opportunity Development			Opportunity Exploitation		
Project Management		**Adaptive Project Based Approaches to Manage the Fuzzy Front End of Entrepreneurial Opportunities**		**Adaptive/Predictive Project Planning to Scale up and/or establishing operations**		
Entrepreneurial Life Cycle						
	Pre-Entrepreneurial Venture			Setting up Your Venture		Growing Your Business
	ideation	Concepting	Commitment	Validation	Scaling	Establishi

Inspired from http://www.startupcommons.org/

Pre Lecture Reading: Dealing with a Project's "Fuzzy Front End".

Discussion Points:

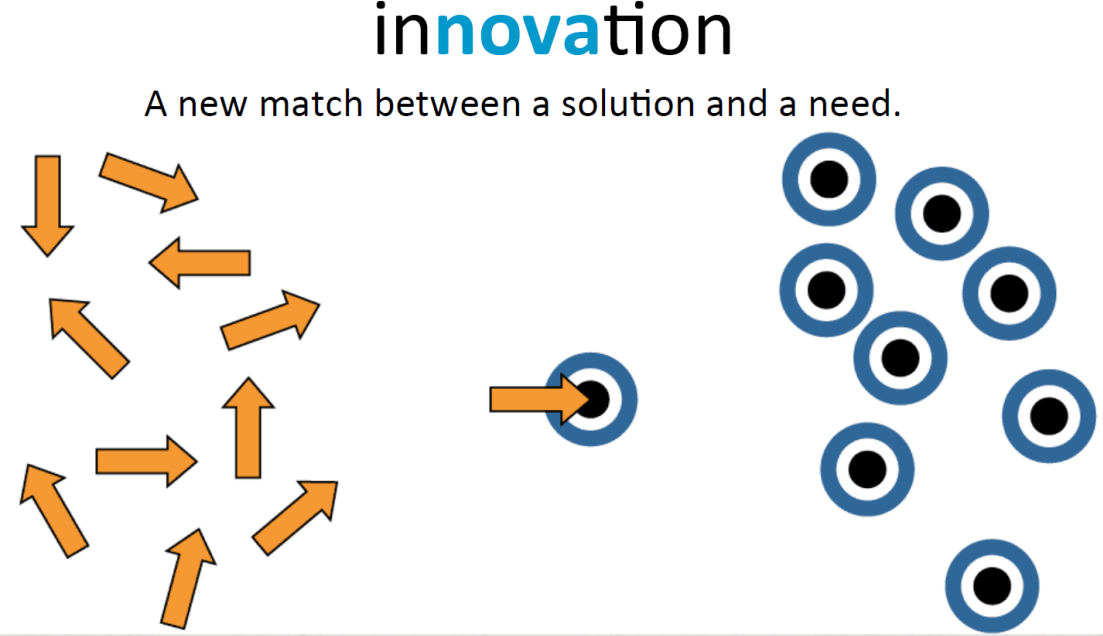
1. What do we mean by the fuzzy front end of a project?
2. What types of projects are likely to have more 'fuzziness' at their initiation?
3. How the 'fuzziness' associated with the front end of projects can be managed?

Fuzzy front end of innovative projects

The new product development process can be broken down in several ways, many of which often include the following phases/stages*:

1. **Fuzzy front end (FFE)** is the set of activities employed before the more formal and well-defined requirements specification is completed. requirements speak to what the product should do or have, at varying degrees of specificity, to meet the perceived market or business need.
2. **Product Design Features/Product scope:** the development of both the high-level and detailed-level design of the product: which turns the what of the requirements into a specific how this particular product will meet those requirements. Validation is also required to confirm that the product features are technically and financially feasible and meet the market requirements.
3. **Project implementation:** the scope of work (project scope) required to deliver the value proposition to the target customer segment.

*Adapted from Kezner (2023, p 134) *Innovation Tools and Processes in Innovation Project Management*



Creativity: The ability to produce work that is useful and novel.
Individual or team creativity is a function of:

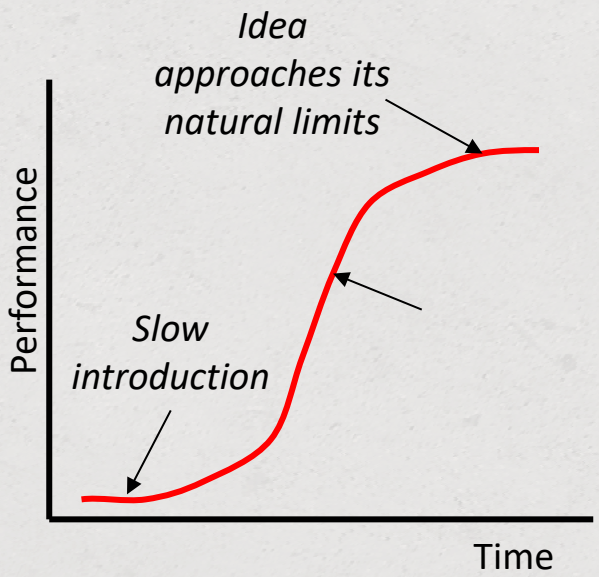
- Intellectual abilities (for example, ability to articulate ideas).
- Knowledge (for example, understand field, but not wed to paradigms).
- Personality (for example, confidence in own capabilities).
- Motivation (for example, rely on intrinsic motivation).
- Environment (for example, support and rewards for creative ideas).

Innovation is the implementation of creative ideas into some useful new device or process.

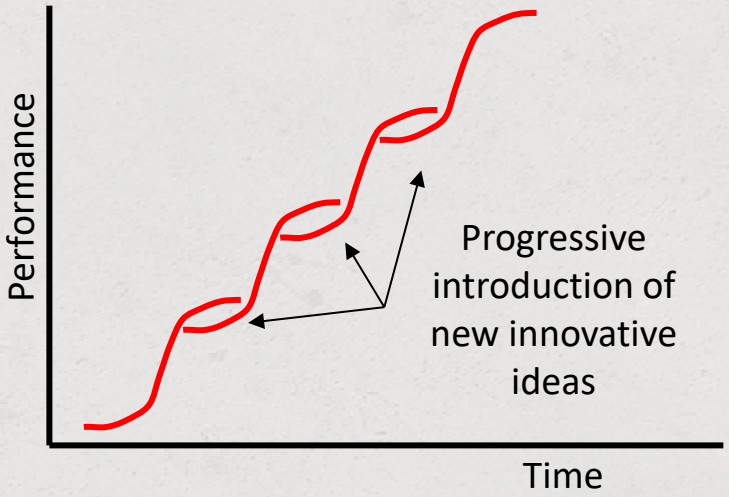
- "Innovation is a process that combines discovering an opportunity, blueprinting an idea to seize that opportunity, and implementing that idea to achieve results. Remember — no impact, no innovation."
- For an idea to be innovative, it must also be useful. Creative ideas don't always lead to innovations because they don't necessarily produce viable solutions to problems.

The Innovation Landscape & Project Management

The S-shaped curve of innovation



(a) Breakthrough Innovation:
The basic S-shaped improvement in performance



(b) Incremental Innovation
Following multiple S-:shaped curves

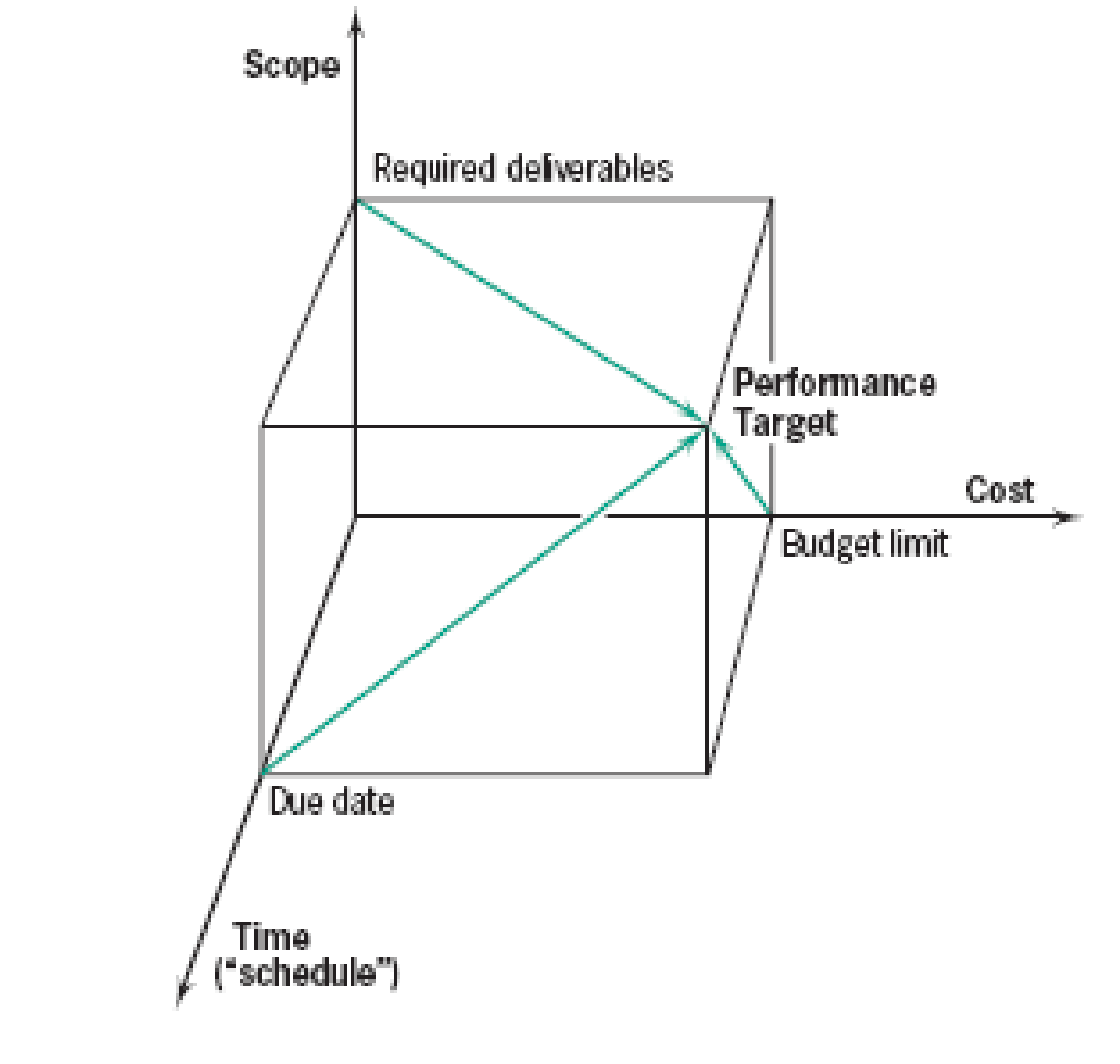
- Radical or breakthrough Innovation: Experiments with new ideas, schemes and approaches to **create entirely new technologies and markets**.
  - Radical advances that may profoundly alter the basis of competition in an industry

- Incremental Innovation: Exploits the existing knowledge base and proven solution/technologies to address the **current customer demands**
  - Small improvements in existing products services

Project Performance Dimensions – Framework 2

Scope
Cost
Time

Project Iron Triangle



Project Success

PROJECT SUCCESS

- **Meeting Agreements**
  —Cost, schedule, and specifications met
- **Customer's Success**
  —Needs met, deliverables used, customer satisfied
- **Performing Organization's Success**
  —Market share, new products, new technology
- **Project Team's Success**
  —Loyalty, development, satisfaction

Source: Adapted from Timothy J. Kloppenborg, Debbie Tesch, and Ravi Chinta, "21st Century Project Success Measures: Evolution, Interpretation, and Direction," *Proceedings, PMI Research and Education Conference 2012* (Limerick, Ireland, July 2012).

Framework 3: Project Governance-Life Cycle

Unit of Analyses

Organization

Project Governance

Organizing projects to maximize benefits by utilizing the limited organizational resources

Unit of Analyses

Individual Project

Project Life Cycle

Initiating, Planning, Performing, Closing

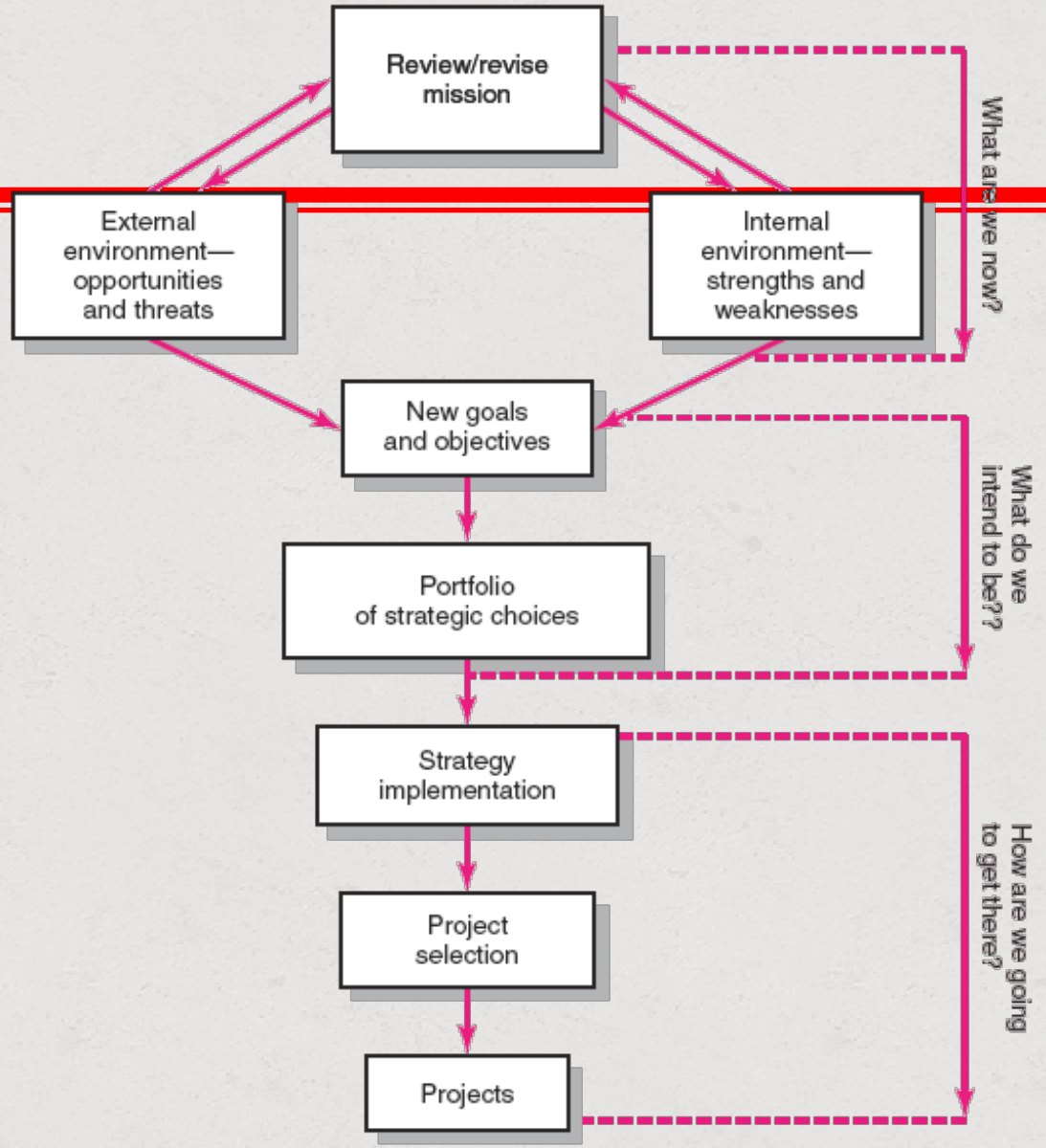
Strategic Management Process

Review and Define

Set long range goals/ objectives

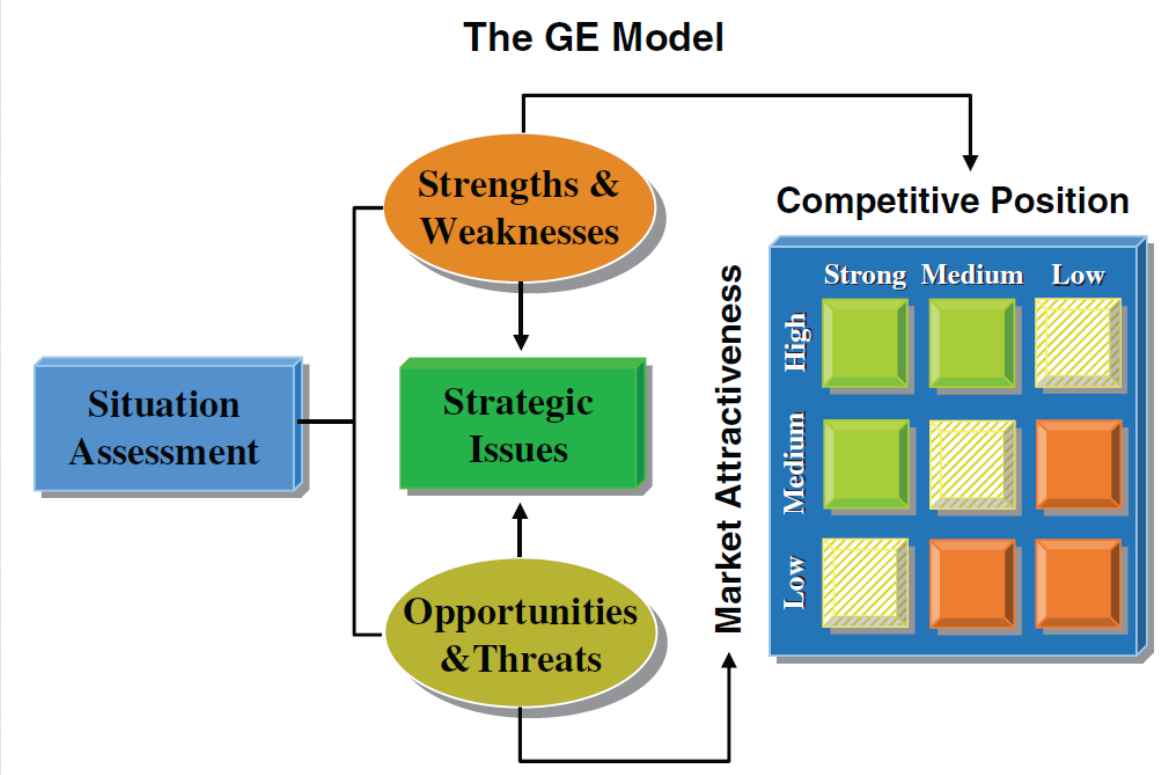
Analyse and formulate strategies

Implement strategies through projects



Project Portfolio Management

- A portfolio or strategic project management office (PMO). The strategic PMO is responsible for participating in the portfolio selection of projects
- Boston Consulting Group (BCG) Project Portfolio*
  - Linking projects to strategy
- GE Portfolio Classification Model*



*Adapted from Kezner (2023, p 77-78) Innovation & Strategic Planning in Innovation Project Management

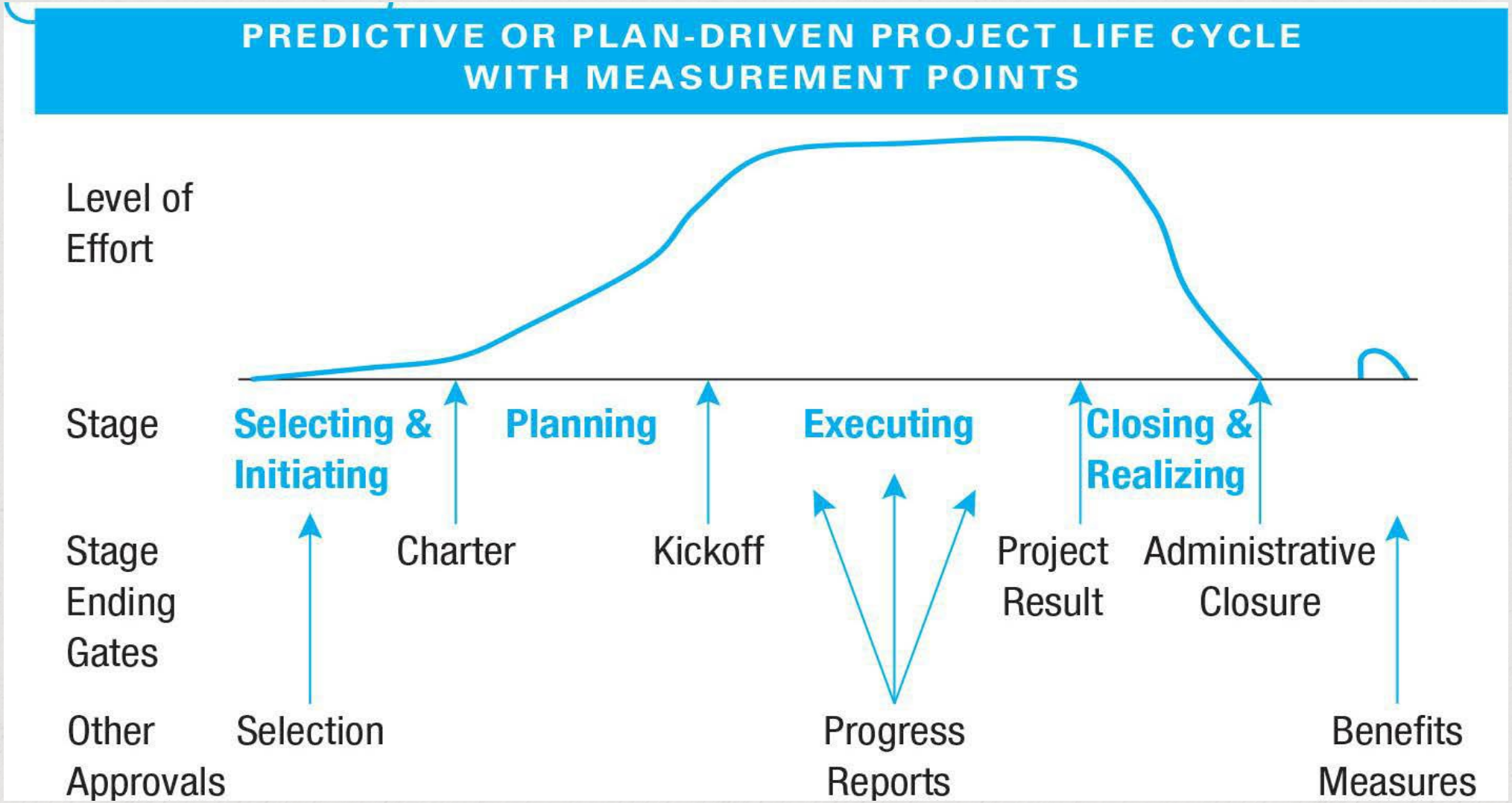
The Assisted Living Project (ALP)

Discussion Point:

1. Plot ALP on Framework 1: Project-Operations Spectrum Framework?
2. What are the next steps for Fred to get the work started on this project? (Hint: identify the current stage of the predictive project life cycle for ALP? words,

Adobe Acrobat Document

Framework 4: The Predictive Project Life Cycle



Predictive extreme → → → → → → → waterfall

Bank of Geelong Mobile App

- Bank of Geelong (BOG) has ambitious plans for expansion and it is aiming to be a leader in the online banking services. BOG has decided to significantly enhance its current mobile banking app so that it is a standout in the Australian banking sector. More specifically, BOG management wants its mobile app to have features that are not currently being offered by any other bank. The President of BOG tasked the Vice-President of Information Systems, Stacey Thomas, to deliver on this broad objective with specific instructions on leaving 'no stone unturned' to have the best mobile app in the industry. One of Stacey's first tasks was to select the project manager. Because of the strategic importance of the project, she had a strong preference for staffing the project internally as opposed to employing the services of one of the many consulting firms available that specialize in these types of projects.

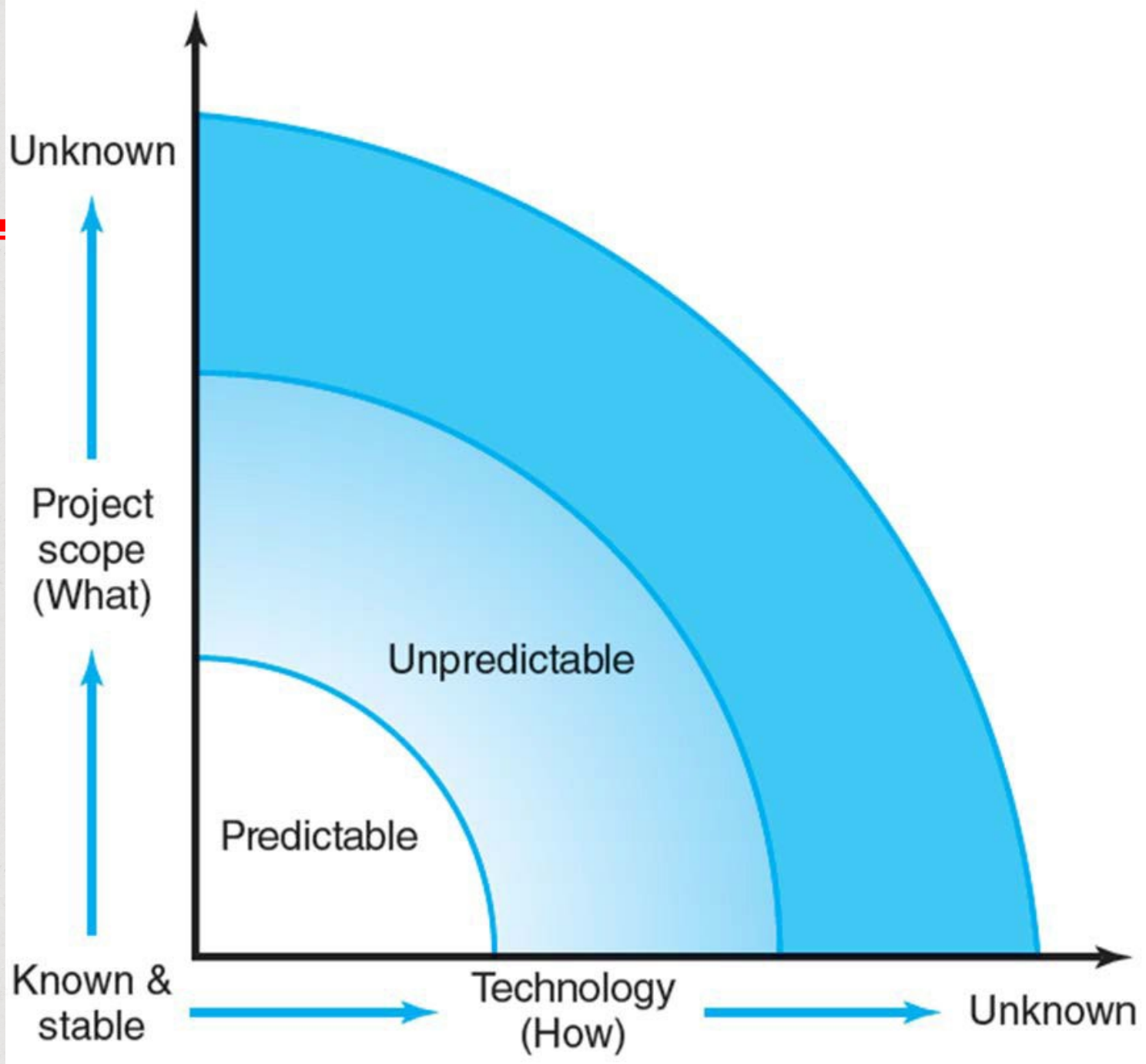
Discussion Points:

1. Compare the Bank of Geelong Mobile App Project with the ALF Project? List down all similarities and differences?

2. If you were Stacey, how would you approach this project?

Framework 5: Project Uncertainty Framework

Larrson & Gray (2020)



http://agilemanifesto.org/

Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

While there is value in the items on the right, we value the items on the left more.

SWINBURNE UNIVERSITY OF TECHNOLOGY

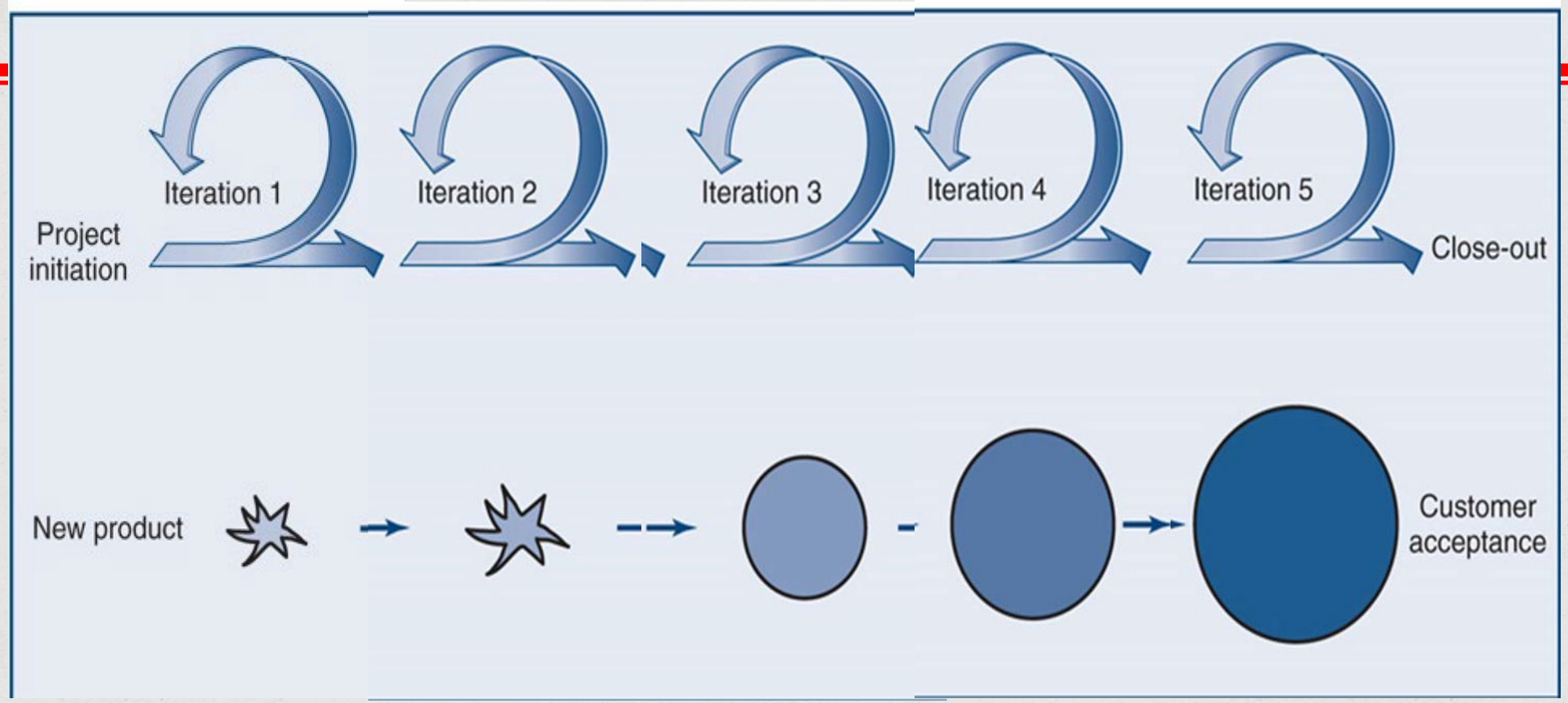
Scrum Methodology - Most Common Agile Method

"Scrum evokes the notion that it's a cooperative effort

Software development is a cooperative game*..."

*http://alistair.cockburn.us/Cooperative+game+manifesto+for+software+development

Incremental, iterative product development



Traditional vs Agile project management

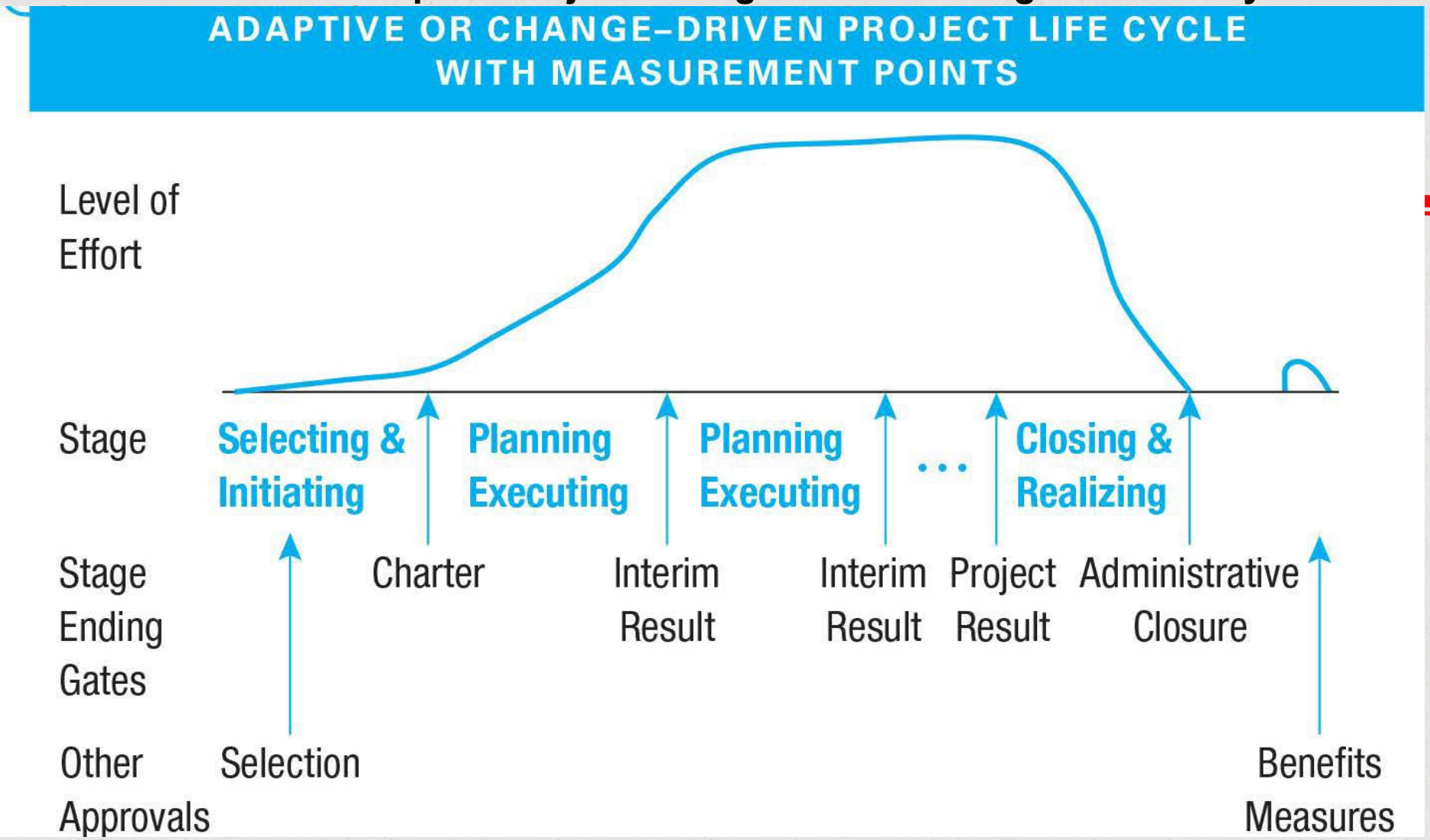
Traditional	Agile
Design up front	Continuous design
Fixed scope	Flexible
Deliverables	Features/requirements
Freeze design as early as possible	Freeze design as late as possible
Low uncertainty	High uncertainty
Avoid change	Embrace change
Low customer interaction	High customer interaction
Conventional project teams	Self-organised project teams

Framework 6: The Right Conditions for Agile PM*

CONDITIONS	FAVORABLE	UNFAVORABLE
MARKET ENVIRONMENT	Customer preferences and solution options change frequently.	Market conditions are stable and predictable.
CUSTOMER INVOLVEMENT	Close collaboration and rapid feedback are feasible. Customers know better what they want as the process progresses.	Requirements are clear at the outset and will remain stable. Customers are unavailable for constant collaboration.
INNOVATION TYPE	Problems are complex, solutions are unknown, and the scope isn't clearly defined. Product specifications may change. Creative breakthroughs and time to market are important. Cross-functional collaboration is vital.	Similar work has been done before, and innovators believe the solutions are clear. Detailed specifications and work plans can be forecast with confidence and should be adhered to. Problems can be solved sequentially in functional silos.
MODULARITY OF WORK	Incremental developments have value, and customers can use them. Work can be broken into parts and conducted in rapid, iterative cycles. Late changes are manageable.	Customers cannot start testing parts of the product until everything is complete. Late changes are expensive or impossible.
IMPACT OF INTERIM MISTAKES	They provide valuable learning.	They may be catastrophic.

Rigby, Darrell K., Jeff Sutherland, and Hirotaka Takeuchi. "Embracing agile." *Harvard Business Review* 94.5 (2016): 40-50.

Framework 7: The Adaptive Project Management to manage uncertainty



Adaptive extreme → → → → → → → *agile*

The Innovation Landscape & Project Management

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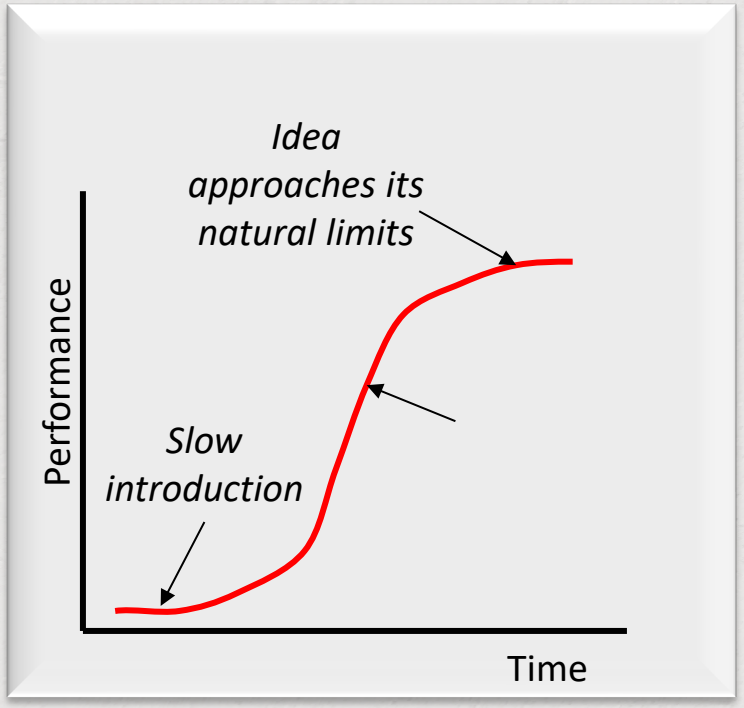

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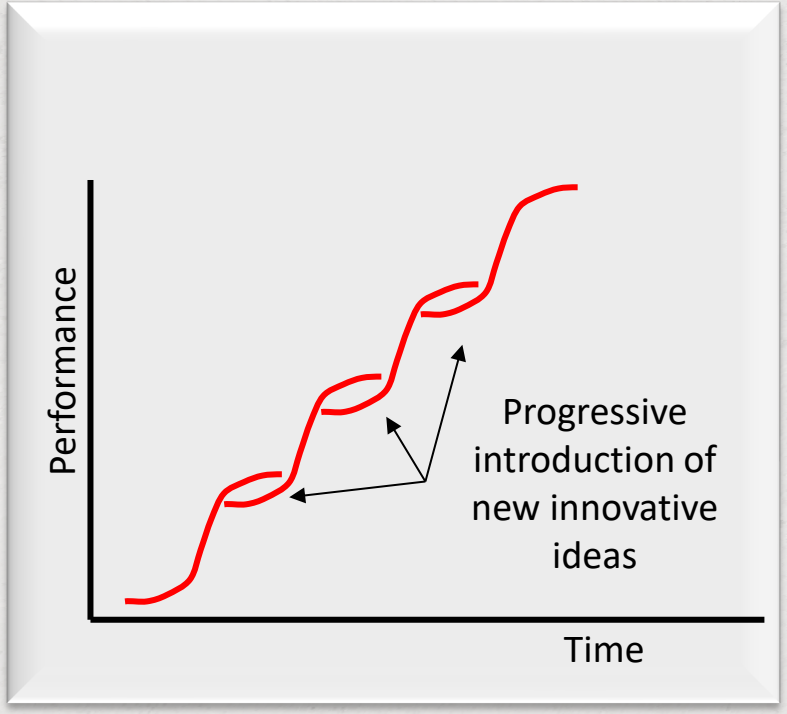

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Revisiting Types of Projects (Davies and Brady 2016)

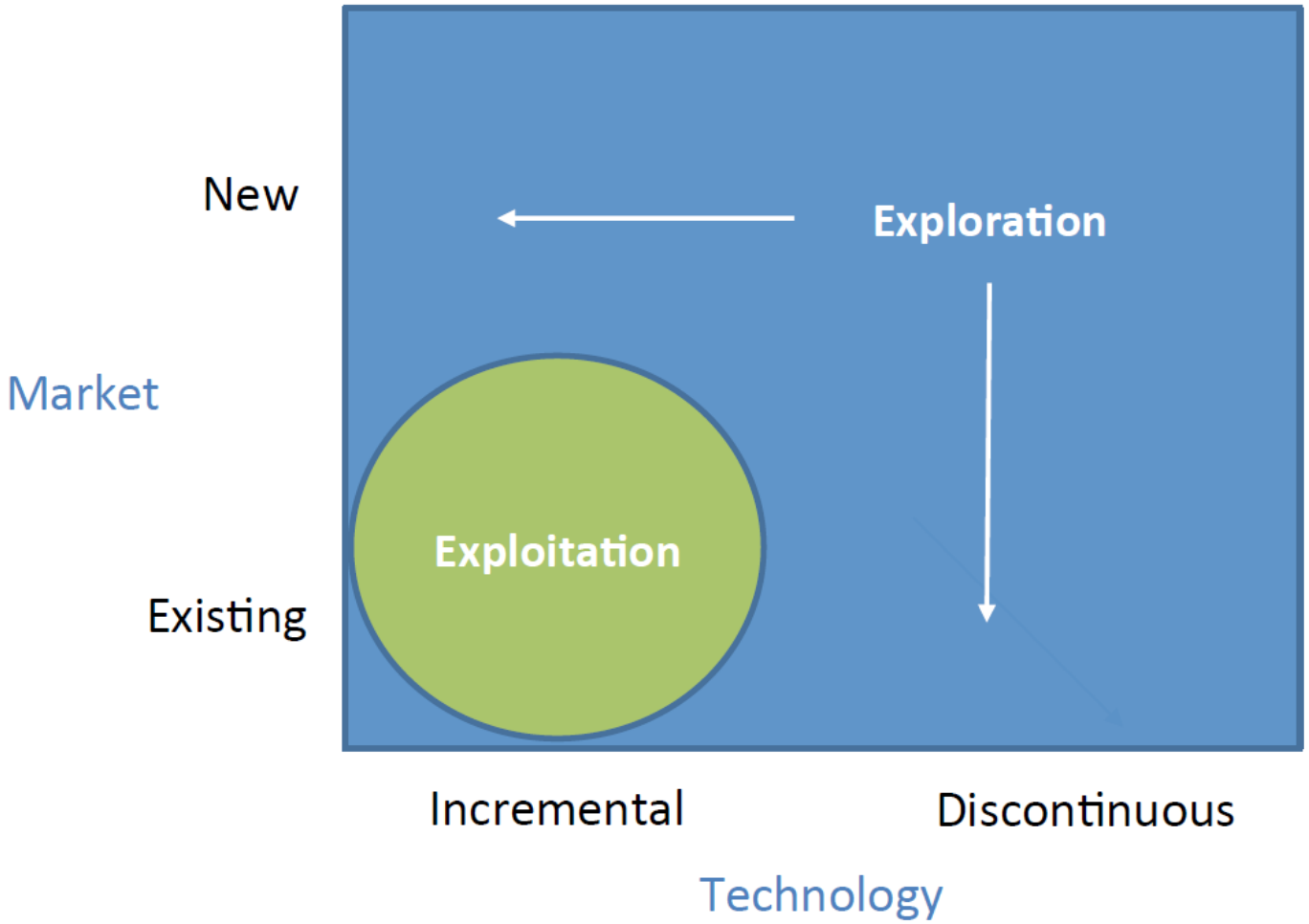
- Classifying by understanding of project scope (Week 1)
  - **Routine Projects**: which *exploit the existing base*, utilise proven technologies and mature products, and address current customer demands;
  - **Innovative Projects:** 'which support *base-moving strategies* that explore innovative alternatives, *experiment* with new ideas, schemes and approaches, and create *entirely new technologies and markets.*

Framework 1: Project –Operations Spectrum

- Bringing together Framework 1 with Davies and Brady (2016) Project Typology
  - Innovative Projects : Towards the right side of the spectrum
    - Focussed on ***exploration*** and characterized by a high degree of novelty and complexity
    - Suitable for exploration strategy and generally, implemented through adaptative project management approach (Framework 7)
  - Routine Projects: Towards the left side of the spectrum
    - focused on ***exploitation*** and organised to achieve predefined goals with a given set of resource constraints (time, cost and quality).
    - Suitable fo exploitation strategy and generally implemented through traditional project management (Framework 4)

The Innovation Landscape and the strategies to achieve innovative outcomes

- **Exploration* -** "a pursuit of new knowledge" or "search, variation, risk-taking, experimentation, flexibility, discovery, and innovation."
- **Exploitation* -** "the use and development of things already known" or "scale and scope of knowledge created or acquired relative to an organization's existing knowledge base."
- **Exploration – Exploitation*** *more of continuum than a dichotomy i.e firms may transition between the two states. In other words, exploration may evolve into exploitation.*
- **Organizational Ambidexterity:** When an organization relies on both exploitation and exploration to create value i.e Ambidexterity exploit a firm's current routines, whilst exploring new opportunities that will define the future (Davies and Brady 2016)



*Lavie, Dovev, Uriel Stettner, and Michael L. Tushman. "Exploration and exploitation

- **Organizational Ambidexterity*:** When an organization relies on both exploitation and exploration to create value
  - Exploitative business = core business activities – more towards repetitive outcomes on framework 1
  - Exploratory business – base moving activities – more towards unique outcomes in framework 1

Alignment of:	Exploitative Business	Exploratory Business
Strategic intent	cost, profit	innovation, growth
Critical tasks	operations, efficiency, incremental innovation	adaptability, new products, breakthrough innovation
Competencies	operational	entrepreneurial
Structure	formal, mechanistic	adaptive, loose
Controls, rewards	margins, productivity	milestones, growth
Culture	efficiency, low risk, quality, customers	risk taking, speed, flexibility, experimentation
Leadership role	authoritative, top down	visionary, involved

The ambidextrous organization*

Framework 1: Project –Operations Spectrum

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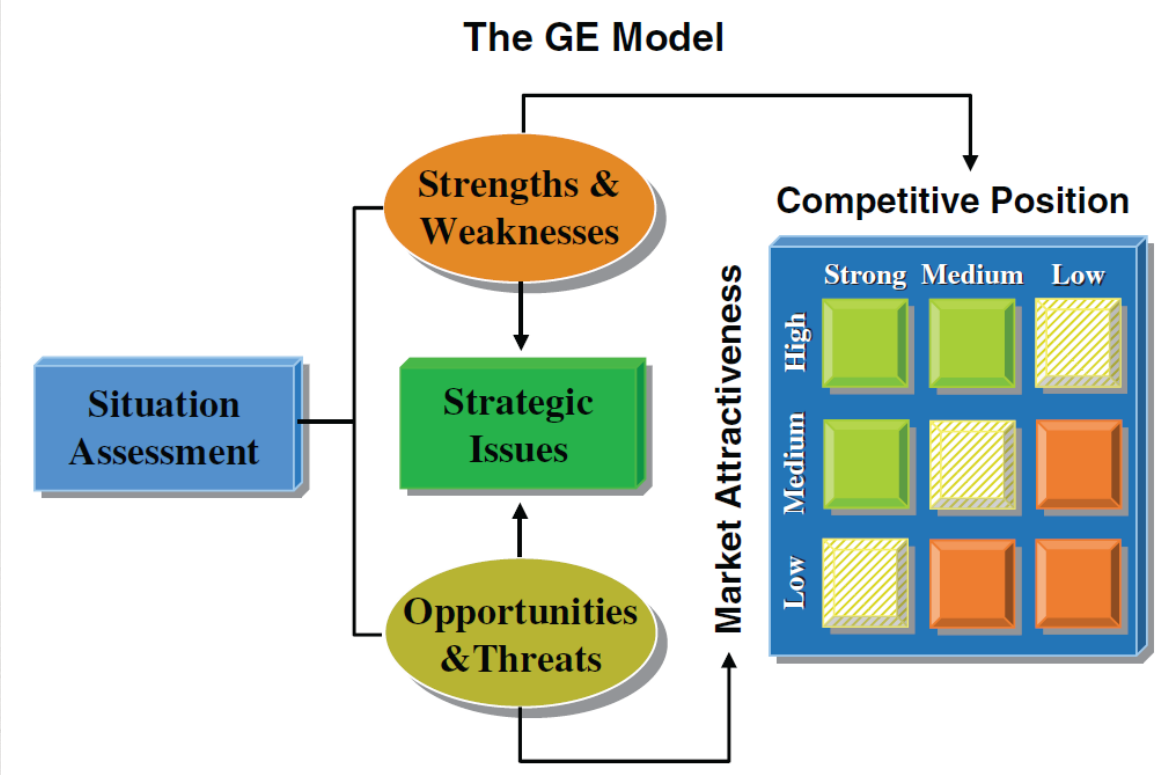

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Lecture Review

Learning Outcomes:

LO1. Understanding the project performance dimensions

LO2. Familiarization with the Project Governance – Life Cycle Framework

LO3. Basic understanding of tasks initiated in different stages of project life cycle

LO4. Appreciating the difference between predictive and adaptive project management and the decision frameworks to select the right project management approach.

References

- Davies, A. , & Brady, T. (2016). Explicating the dynamics of project capabilities. *International Journal of Project Management, 34* , 314–327
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